NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-39035/4E
31 January 1994
SUPERSEDING
MIL-R-39035/4D
28 September 1979

PERFORMANCE SPECIFICATION

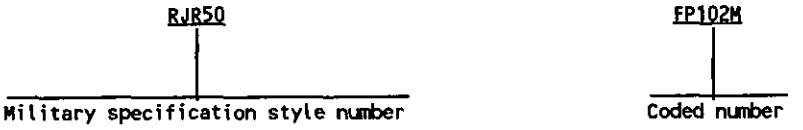
RESISTORS, VARIABLE, NONWIRE-WOUND
(ADJUSTMENT TYPE, SINGLE TURN),
ESTABLISHED RELIABILITY
STYLE RJR50

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RJR50, established reliability, adjustment type, single turn, nonwire-wound, variable resistors. This style is available in characteristic F and H.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification shall be identified by a PIN which shall consist of a basis style of this specification and a coded number. The PIN shall be in the following form:




2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-39035 - Resistors, Variable, Nonwire-wound (Adjustment Type), Established Reliability, General Specification for.

(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:58Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-39035, and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/4 watt for characteristics F and H.

3.4 Terminals. Characteristics F and H are available in P- type terminal.

3.5 Nominal resistance value and maximum rated ac or dc working voltage. Nominal resistance values and maximum rated ac or dc working voltages shall be as specified in table I.

3.6 Actual effective electrical travel. Actual effective electrical travel shall be 215° minimum.

3.7 Torque.

3.7.1 Operating. Operating torque shall be a maximum of 2.0 ounces-inches.

3.7.2 Stop. The stop shall withstand a force of 4 ounce-inches when this amount of torque is applied through the operating shaft to the stop.

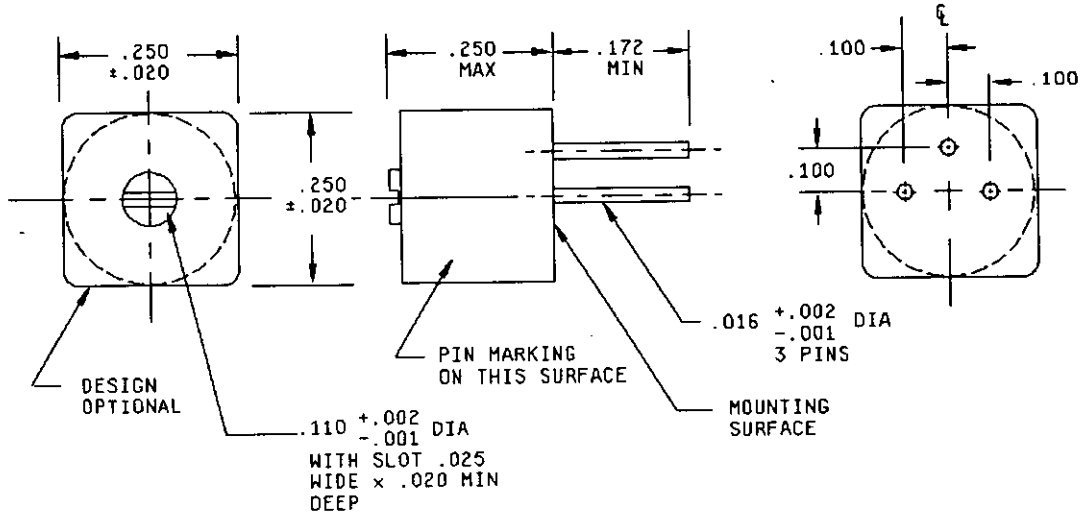
3.8 Maximum voltage. Maximum rated ac or dc working voltage shall be 200 volts.

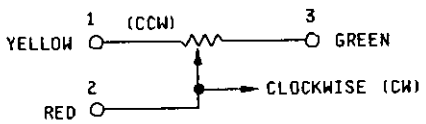
TABLE I. Nominal resistance value and maximum rated ac or dc working voltage.

| Nominal resistance value | Maximum rated ac or dc working voltage per characteristics |
|---|---|
| | F and H |
| Ohms | |
| 10 | 1.58 |
| 20 | 2.23 |
| 50 | 3.54 |
| 100 | 5.0 |
| 200 | 7.07 |
| 500 | 11.1 |
| 1,000 | 15.8 |
| 2,000 | 22.3 |
| 5,000 | 35.4 |
| 10,000 | 50.0 |
| 20,000 | 70.7 |
| 25,000 | 79.0 |
| 50,000 | 111 |
| Megohms | |
| 0.10 | 158 |
| 0.20 | 200 |
| 0.25 | 200 |
| 0.50 | 200 |
| 1.00 | 300 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:58Z
Check the source to verify that this is the current version before use.




SCHMETIC DIAGRAM

| Inches | mm |
|---|---|
| .001 | 0.03 |
| .010 | 0.25 |
| .015 | 0.38 |
| .016 | 0.41 |
| .020 | 0.51 |
| .025 | 0.64 |
| .100 | 2.54 |
| .110 | 2.79 |
| .250 | 6.35 |
| .312 | 7.92 |
| .375 | 9.53 |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
3. Metric equivalents are given for general information only.
4. Mounting means are by use of pins only.
5. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
6. The head of the actuating screw may or may not be flush with or recessed in the body.

FIGURE 1. Style RJR50 resistors.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:58Z
Check the source to verify that this is the current version before use.

3.9 Marking. Due to size limitations, this style resistor shall be marked with the following minimum information:

7523A - Date code, lot code.
FP102MJ - Characteristic, terminal, resistance value, failure rate and JAN marking.
12345 - Source code.

The complete marking is required on the unit package.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39035, and as specified herein.

4.2 Dielectric withstanding voltage. The magnitude of test voltage shall be 600 volts rms at atmospheric pressure, and 250 volts rms at reduced barometric pressure.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-R-39035 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.(see 2.1).

6.3 Weight. The weight is .021 ounces (0.6 grams).

6.4 Application of packaging

6.4.1 Shipments to Government activities. The packaging requirements of this specification are primarily intended for the preparation of resistors for shipment to Government activities.

6.4.2 Shipments of ER (resistors) to Government contractors and manufacturer's distributors. Federal Standard No. 356, Commercial Packaging of Supplies and Equipment, should be specified to Government contractors and subcontractors and to category A and B distributors as defined in MIL-STD-790. The marking of the unit container should not be required provided the intermediate container are marked.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:58Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, EC, OS, MC
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5905-1367)



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:58Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER MIL-R-39035/4E | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|

**3. DOCUMENT TITLE**
RESISTORS, VARIABLE, NOWIRE-WOUND (ADJUSTMENT TYPE, LEAD-SCREW ACTUATED), ESTABLISHED RELIABILITY STYLE RJR50

**4. NATURE OF CHANGE** (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, first, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME<br>645 LOG/LGMEP | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>513-296-6696 | (2) AUTOVON<br>986-6696 |
|---|---|---|
| c. ADDRESS (Include Zip Code)<br>Gentile Station<br>1060 Hamilton Street<br>Dayton, OH 45444-4500 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:58Z
Check the source to verify that this is the current version before use.